



SERA Cell Morphology Competition Pre-Proposal Assignment

Introduction

Research and present a pre-proposal for the SERA Cell Morphology Competition.

You should: (a) select one human cell line meeting all contest constraints and (b) nominate **three** immunofluorescence morphological markers.

High quality submissions will be featured on the UATX Astronautics Substack.

Cell line contest constraints:

- Adherent
- 2-D growth
- Biosafety Level 1 (BSL-1)

Requirements

Section	Content to Include	Suggested Length
1. Cell Line Description	• Name and catalog number (from the approved list) • Cell type, tissue of origin, unique features (e.g., cancer, CRISPR-edited) • Prior use in microgravity or related research (cite studies) • Justification: Why is this line ideal for detecting morphological change in microgravity? Consider adhesion strength, cytoskeletal features, flight heritage, and relevance to human spaceflight.	≤ 500 words
2. Morphological Markers (three subsections, one per marker)	For each marker provide a table—or concise paragraphs—covering: • Marker name & catalog link • Normal cellular function / structure labeled • Relevance to microgravity: what change do you predict and why? • Brief summary of previous research using this marker (space or Earth) • Expected readout/outcome in your experiment	~200 words per marker
3. Integrated Rationale	Synthesize how the chosen cell line and marker set together will yield a novel insight..	≤ 300 words



Grading Rubric

Criterion	Pts
Adherence to all competition requirements (BSL-1, 2-D, adherent; three qualified markers)	5
Scientific justification of cell line	10
Scientific justification of each marker (3 × 5 pts)	15
Clarity and Organization	10
Total: 40 pts	

Submission Instructions

- Submit your pre-proposal as a word document or PDF on Populi
- Deadline: **11:59pm Fri June 6, 2025**

Cell Line Catalogs

Select cell lines from the following catalog options:

 Cell Line Options

Morphology Markers Catalog:

<https://www.thermofisher.com/antibody/primary/query/IF>